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BBO-MARNet: Bayesian Band Optimization and Mask-Aware Reconstruction Network for Efficient Hyperspectral Anomaly Detection

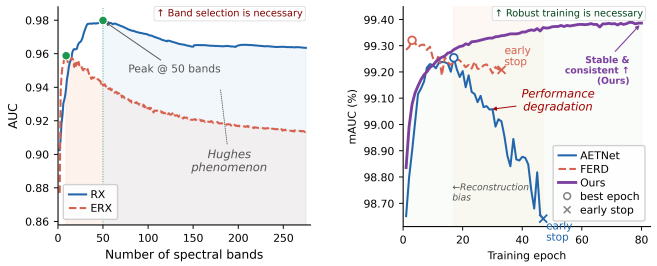
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***Abstract*—Hyperspectral anomaly detection is essential for blind target discovery, yet its effectiveness is hindered by high spectral redundancy and target-background entanglement. Existing reconstruction-based methods rely on task-agnostic full-spectrum modeling, which lacks discriminative subspace alignment and often leads to anomaly memorization. To address this limitation, we propose the Bayesian Band Optimization and Mask-Aware Reconstruction Network (BBO-MARNet). The framework first employs Bayesian optimization to identify a compact, detection-oriented band subset, reducing spectral redundancy prior to feature learning. A lightweight spatial-spectral masked autoencoder is then applied to reconstruct missing information from contextual cues, mitigating anomaly memorization. Additionally, a tripartite composite loss function, integrating multi-scale reconstruction, gradient consistency, and internal regularization, is designed to enhance background representation and preserve structural boundaries. Experiments on five benchmark datasets, including a newly curated maritime dataset, demonstrate that BBO-MARNet consistently outperforms state-of-the-art methods while maintaining high efficiency with only 1.69M parameters. The source code is available at https://github.com/Sonel-YXL/BBO-MARNet.**

***Index Terms*—Hyperspectral anomaly detection, band selection, Bayesian optimization, masked autoencoder, self-supervised learning, lightweight neural network.**

I. Introduction

HYPERSPECTRAL imaging (HSI) captures continuous spectral responses across hundreds of narrow bands, offering an unparalleled capability for fine-grained material characterization [1]. In critical remote sensing operations—spanning maritime hazard monitoring to defense surveillance—targets of interest (e.g., maritime vessels, camouflaged aircraft) frequently exhibit subtle spectral deviations from highly complex and heterogeneous backgrounds [2]. Hyperspectral anomaly detection aims to isolate these uncharacterized deviations without requiring pre-specified target spectra or target-scene labels during model training and inference. Crucially, in real-world scenarios, anomalies are fundamentally defined by their unique *material compositions*—such as the specific metallic alloys of an aircraft or the synthetic coatings of a ship—rather than purely spatial or structural irregularities [3]. In this work, target-scene labels are not used during target-scene training or inference, while the offline AUC-based band calibration uses ground truth from a reference scene containing similar materials. Consequently, the core challenge of HAD lies not merely in feature extraction, but in identifying a task-relevant spectral subspace that is maximally sensitive to discriminative material signatures.



(a) Hughes phenomenon. (b) Identity mapping dilemma.

Fig. 1. Two fundamental bottlenecks in contemporary HAD and our corresponding motivation. (a) The Hughes phenomenon caused by spectral redundancy, highlighting the necessity of task-aware band selection. (b) The identity mapping dilemma in full-spectrum reconstruction, which motivates our mask-aware reconstruction strategy to prevent reconstruction bias.

Despite significant advancements, contemporary HAD paradigms remain hindered by the intrinsic high dimensionality and severe spectral redundancy of hyperspectral data [4]. Traditional and deep learning-based approaches alike typically operate under a full-spectrum modeling assumption, treating all contiguous bands uniformly in a task-agnostic manner during feature extraction. However, this indiscriminate processing is fundamentally suboptimal. In reality, the critical features distinguishing an anomalous target from a background manifold are often sparsely encoded within a narrow subset of bands. Processing the entire spectral cube not only incurs prohibitive computational overhead but actively dilutes these material-sensitive features, exacerbating the Hughes phenomenon (as visually motivated in Fig. 1(a)) [5]. Therefore, HAD should be fundamentally reformulated as a *material-aware and task-driven subspace learning* problem.

To circumvent the limitations of traditional mathematical simplifications, the field has experienced a decisive paradigm shift toward data-driven deep learning architectures, particularly reconstruction-based models (e.g., Autoencoders, GANs, and Transformers) [3, 6]. These frameworks operate on the premise that anomalies—being statistically rare—cannot be faithfully reconstructed by a network optimized on background priors; thus, substantial reconstruction residuals serve as definitive indicators of anomaly presence [7]. However, this prevailing paradigm suffers from a fundamental and structurally induced

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limitation, namely the "identity mapping" dilemma [8]. When exposed to the full, redundant spectral space, highly parameterized networks frequently discover spectral "shortcuts," trivially mapping inputs to outputs without genuinely modeling the intrinsic background manifold. This reconstruction bias forces the network to inadvertently reconstruct anomalous targets alongside the background (Fig. 1(b)), severely diminishing the reconstruction discrepancy essential for detection. More critically, this failure is not incidental but intrinsically linked to spectral redundancy, where highly correlated bands provide excessive degrees of freedom that enable the network to bypass meaningful representation learning.

To systematically dismantle these intertwined bottlenecks, we propose a novel and unified framework: the Bayesian Band Optimization and Mask-Aware Reconstruction Network (BBO-MARNet). We argue that resolving the identity mapping dilemma and overcoming spectral redundancy are not isolated engineering tasks, but two facets of the exact same objective. BBO-MARNet tightly couples subspace discovery with conditional reconstruction dynamics, ensuring that subspace discovery and reconstruction dynamics are explicitly co-optimized to maximize anomaly-background discriminability rather than generic reconstruction fidelity.

Specifically, the Bayesian Band Optimization (BBO) module functions as *an explicit task-driven material-aware subspace discovery mechanism*. Moving beyond heuristic, task-agnostic dimensionality reduction [9], BBO formulates band selection as a data-driven combinatorial optimization problem, dynamically isolating the minimal band subset that directly optimizes anomaly detection objectives by maximizing anomaly-background separability. Built explicitly upon this optimized subspace, the Mask-Aware Reconstruction (MAR) module introduces a *conditional reconstruction mechanism*. By deliberately masking spatial-spectral patches, the network is strictly compelled to hallucinate missing semantics from surrounding contextual priors under incomplete observations. Crucially, because this masked inference occurs within a strictly non-redundant subspace, the network is entirely deprived of the spectral shortcuts that enable identity mapping.

The BBO module determines the discriminative spectral subspace, while the MAR module governs the reconstruction dynamics to explicitly amplify anomaly-background discrepancies. By conceptually elevating HAD from indiscriminate spatial-spectral processing to target-driven material subspace learning, BBO-MARNet achieves exceptional detection performance.

The main contributions of this article are summarized as follows:

- **Task-Aware Spectral Subspace Modeling:** We reformulate hyperspectral anomaly detection as a task-driven subspace discovery problem, where a Bayesian band optimization (BBO) strategy dynamically identifies a minimal yet discriminative spectral subset, explicitly mitigating redundancy and enhancing target separability.
- **Mask-Guided Spatial-Spectral Reconstruction Mechanism:** We introduce a masking-based spatial-spectral reconstruction paradigm that enforces context-aware inference, fundamentally overcoming the identity mapping issue inherent in conventional full-spectrum reconstruction methods.
- **Structure-Preserving Background Modeling Objective:** We propose a tripartite loss function that jointly enforces background reconstruction fidelity, gradient-level structural consistency, and internal regularization, enabling effective background manifold disentanglement while preserving anomaly-sensitive boundaries.

The remainder of this article is organized as follows. Section II reviews the related literature on hyperspectral band selection and deep anomaly detection. Section III systematically details the proposed BBO-MARNet framework. Section IV delineates the experimental setup, datasets, and evaluation metrics. Section V presents a comprehensive analysis of the detection results and ablation studies. Finally, Section VI concludes the article.

II. Related Work

A. Hyperspectral Anomaly Detection

Traditional hyperspectral anomaly detection methodologies have heavily relied on statistical background modeling, epitomized by the Reed-Xiaoli (RX) detector and its variants [10, 11]. While computationally elegant, these approaches inherently assume specific multivariate background distributions (e.g., Gaussian or elliptically symmetric). Consequently, they frequently struggle to characterize the non-homogeneous and highly complex background clutter typical of real-world scenarios, inevitably leading to elevated false alarm rates [10, 12].

Beyond statistical detectors, representation- and tensor-based methods provide model-driven alternatives for HAD. Representation-based approaches typically characterize the background through low-rank or subspace models and identify anomalies from sparse components or reconstruction residuals, including low-rank sparse separation, collaborative sparsity, and learnable background-endmember representations [7, 13, 14]. Tensor-based methods further preserve the intrinsic spatial-spectral structure of HSIs and exploit multilinear low-rank and structural priors through tensor ring or layered tensor decomposition [15, 16]. Although these methods improve background and structural modeling, they generally rely on predefined priors and iterative optimization and do not explicitly perform task-driven discrete band selection. This motivates a lightweight pipeline that first identifies an informative spectral subset and then learns a compact reconstruction model in the optimized subspace.

To circumvent the limitations of rigid statistical assumptions, recent literature has experienced a decisive paradigm shift toward data-driven deep learning frameworks [10]. Grounded in the premise that rare and structurally anomalous targets cannot be faithfully reconstructed by models optimized exclusively on background priors, representation learning paradigms have rapidly evolved. Standard Convolutional Neural Networks (CNNs) have been increasingly augmented or superseded by advanced architectures designed to capture complex spatial-spectral dependencies. For example, Graph Autoencoders (GAEs) leverage non-Euclidean geometries to model long-range

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background dependencies and incorporate low-rank properties, thereby enhancing model transferability [12, 17]. Concurrently, Transformer-based architectures explicitly delineate nonlinear hyperspectral mixtures through integrated global and local spatial-spectral contexts [18, 19], while diffusion models conditioned on endmember features have demonstrated efficacy in excavating subpixel anomalies from highly mixed backgrounds [18].

Despite robust feature extraction capabilities, these generative models frequently suffer from the critical "identity mapping" dilemma, inadvertently reconstructing anomalous targets alongside the background. To enforce strict background purification, contemporary studies have introduced specialized spatial masking mechanisms. Architectures such as the Sparse-Masked Convolutional Network (SMCNet) [10], spatial-spectral cross-guided mask autoencoders [11], and sliding dual-window networks [20] impose blind-spot constraints, compelling the inference of central pixels to rely entirely on surrounding contextual priors. However, while these masking strategies effectively prevent target memorization, they predominantly process the entire high-dimensional spectral cube indiscriminately. This exhaustive processing not only imposes a severe computational burden but also forces the network to absorb redundant and noisy spectral channels, ultimately bottlenecking target-background separability. Consequently, mitigating this pervasive spectral redundancy via targeted band selection has emerged as an imperative prerequisite for developing efficient and highly discriminative HAD systems.

B. Band Selection for Anomaly Detection

Despite the profound success of deep generative models in HAD, the inherent high dimensionality of hyperspectral imagery (HSI) precipitates the well-documented "curse of dimensionality" [21, 22]. While the contiguous spectral bands provide rich diagnostic information, this continuity inherently introduces severe inter-band correlation and collinearity. Indiscriminate processing of the full spectral cube not only incurs prohibitive computational overhead but also exacerbates the degradation of feature learning due to the inclusion of uninformative or noisy channels [21].

To mitigate these structural bottlenecks, band selection (BS) has emerged as a crucial dimensionality reduction strategy. The primary objective is to isolate a compact, highly informative spectral subset that preserves the intrinsic data manifold, thereby accelerating computation and enhancing overall detection discriminability [21, 22]. Traditionally, BS operates as an isolated preprocessing step, relying heavily on heuristic statistical criteria such as variance, information entropy, or clustering frameworks [22, 23]. More recently, research efforts have shifted toward tailoring BS explicitly for anomaly detection. For instance, unsupervised methods leveraging out-of-distribution (OOD) modeling and spatial-spectral outlier scores have been proposed to retain band subsets with naturally high target-to-background contrast [22, 24]. Concurrently, task-constrained frameworks, such as BSRegNet, integrate BS prior to deep regularization, empirically demonstrating that actively pruning spectral redundancy substantially benefits downstream background reconstruction and anomaly localization [23].

However, a critical methodological gap persists. Although recent frameworks attempt to couple BS with the detection pipeline, the actual selection mechanisms remain largely heuristic or rely on proxy optimization targets rather than the definitive downstream detection metric (e.g., AUC). Because the combinatorial search space of band subsets is rarely optimized directly against the detector's end performance, there is no guarantee that the retained subset is genuinely task-optimal. Consequently, subtle spectral signatures critical for identifying weak anomalies are frequently inadvertently discarded. This limitation highlights an urgent necessity for a truly task-aware optimization paradigm—one that dynamically orchestrates the dimensionality reduction phase with the ultimate anomaly detection objective, thereby ensuring robust feature representation and adaptability across diverse operational scenarios.

C. Cross-scene Generalization

Although contemporary task-constrained and generative deep learning models achieve state-of-the-art accuracy on specific benchmark datasets, they are frequently hindered by excessive parameterization and dataset-specific overfitting. Leading HAD architectures routinely necessitate tens to hundreds of millions of parameters, rendering them fundamentally impractical for resource-constrained applications, such as real-time processing on edge devices or on-board satellite systems [10, 11, 25]. Consequently, cross-scene generalization remains a critical bottleneck; models tightly fitted to a specific hyperspectral scene frequently exhibit precipitous performance degradation when deployed in novel environments characterized by distinct atmospheric conditions, sensor noise profiles, or heterogeneous topographies [25].

To bridge this generalization gap, recent research has increasingly focused on developing domain-adaptive HAD paradigms. One prominent trajectory formulates anomaly detection as an Out-of-Distribution (OOD) recognition problem. By leveraging graph neural networks and energy-based scoring, these frameworks learn resilient topological structures to maintain detection efficacy across disparate urban and natural scenes [26]. Alternatively, deep unfolding networks explicitly embed classical low-rank representation (LRR) priors into neural architectures, imparting interpretable mathematical constraints that naturally mitigate overfitting compared to pure black-box networks [27]. Furthermore, advancements in cross-domain continual learning, such as CL-BioGAN, employ active parameter forgetting and self-attention strategies to adapt to shifting background distributions without catastrophic forgetting [28].

At the forefront of zero-shot generalization, paradigms like AETNet utilize universal anomaly enhancement networks and image-level training to process novel scenes, ostensibly circumventing the need for extensive parameter fine-tuning [25]. However, a fundamental paradox persists: achieving this robust cross-scene generalization typically comes at the prohibitive cost of immense computational overhead or a heavy reliance on massive pre-training datasets. Therefore, a compelling need

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remains for ultra-lightweight, parameter-efficient architectures capable of autonomous, *in situ* adaptation to novel operational environments. This unaddressed operational imperative directly motivates the proposed BBO-MARNet framework, which synergizes task-aware band optimization with a highly compact mask-guided network to ensure optimal target-background separation and robust generalization across diverse hyperspectral scenes without compromising computational efficiency.

D. Hyperspectral Foundation Models and Recent Architectures

The emergence of hyperspectral foundation models has shifted HSI interpretation from task-specific learning toward transferable representations across scenes, sensors, and downstream tasks. HyperSIGMA introduces a vision Transformer-based hyperspectral foundation model that is scalable to over one billion parameters and is pretrained on the HyperGlobal-450K dataset. Its sparse sampling attention and spectral enhancement module are designed to alleviate spatial-spectral redundancy while supporting diverse high- and low-level HSI tasks [29]. HyperFree further addresses heterogeneous spectral configurations by dynamically constructing input embeddings from a learned full-spectrum weight dictionary, enabling channel-adaptive and tuning-free inference across different HSIs; its effectiveness has been demonstrated on five tasks and eleven datasets [30]. These studies demonstrate the potential of large-scale pretraining and transferable spectral representations. However, hyperspectral anomaly detection differs from large-scale hyperspectral representation learning, as it focuses on discovering unknown targets through background modeling and anomaly–background separation rather than predefined category recognition. The present work focuses on compact anomaly-oriented subspace learning.

For HTD and HAD, advances in transferability and background modeling are represented by SelfMTL and D3Former. SelfMTL pretrains global-local spectral contrastive representations and transfers them to new target-detection tasks using only one target-background pair, followed by adaptive spatial-spectral enhancement [31]. As an HTD method, however, it relies on prior target information, whereas HAD is generally formulated without target priors. For HAD, D^{3}Former develops a dual-decoder dual-Transformer reconstruction framework with background purification, anomaly stripping, cross-guided attention, and spatial-spectral/frequency-domain constraints to mitigate equivalent reconstruction of background and anomalies [32]. In contrast, BBO-MARNet explicitly introduces task-aware band optimization before mask-aware background reconstruction.

Beyond HTD and HAD, several architectures provide complementary insights into hyperspectral change detection, efficient long-range modeling, multimodal classification, and cross-scene adaptation. SSTKAN employs spectral-spatial and 3-D Kolmogorov-Arnold networks together with distribution alignment and multiscale enhancement for hyperspectral change detection [33]. DAMamba introduces dynamic adaptive scanning for efficient long-range visual modeling [34], while MMD-MLP combines local-global directional MLPs, multiresolution multiscale features, and LiDAR-guided cross-attention for multimodal HSI classification [35]. DHSNet addresses cross-scene HSI classification through dual-head self-training, class-wise feature alignment, and central feature attention [36]. Collectively, these studies highlight transferable representation learning, spectral-spatial interaction, efficient global modeling, and cross-scene adaptation. BBO-MARNet instead focuses specifically on anomaly-oriented HAD with task-aware spectral subset optimization and compact mask-aware reconstruction.

III. Methodology

A. Overall Architecture

The proposed Bayesian Band Optimization and Mask-Aware Reconstruction Network (BBO-MARNet) is a unified framework formulated to address hyperspectral anomaly detection through a tight synergy of task-aware dimensionality reduction and self-supervised semantic reconstruction. The complete algorithmic pipeline, as illustrated in Fig. 2, naturally unfolds across three sequential stages: Bayesian band selection, mask-guided spatial-spectral reconstruction, and composite loss-driven anomaly detection.

Initially, rather than merely mitigating the Hughes phenomenon or alleviating computational overhead, $\mathbf{X}$ is processed by a task-aware Bayesian optimization module to dynamically isolate a material-sensitive spectral subspace. By navigating the highly redundant combinatorial search space, this module strips away task-irrelevant spectral clutters and captures the intrinsic material signatures that maximize target-background separability, yielding a highly discriminative sub-cube $\mathbf{X}_{\mathcal{S}} \in \mathbb{R}^{H\times W\times K}$ ($K \ll B$).

Subsequently, a spatial masking strategy is applied to $\mathbf{X}_{\mathcal{S}}$, intentionally occluding random regional patches. This operation fundamentally prevents the network from trivially memorizing local pixel intensities, compelling it instead to infer missing content from long-range background contextual priors. The masked data cube is then fed into a hierarchical, U-shaped spatial-spectral Transformer architecture. During the encoding phase, the network progressively downsamples the spatial resolution while expanding the channel capacity. Within each hierarchical stage, feature transformations are governed by an alternating dual-branch attention mechanism: a spectral Transformer branch computes global channel covariance to capture long-range spectral dependencies, while a shifted-window spatial Transformer branch models local and cross-window structural contexts. This synergistic spatial-spectral modeling ensures that the non-linear manifold of the hyperspectral data is comprehensively characterized.

In the decoding phase, bottleneck features are upsampled and fused with high-resolution hierarchical features from the encoder via skip connections to strictly preserve fine-grained spatial boundaries. Driven by a tripartite composite loss function that explicitly penalizes background reconstruction errors, preserves structural gradients, and suppresses internal anomaly representations, the network outputs a high-fidelity background reconstruction cube $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times K}$. Finally, the discrepancy between this optimized background manifold and the input cube is leveraged by standard statistical detectors (e.g., the RX algorithm) to compute the ultimate anomaly

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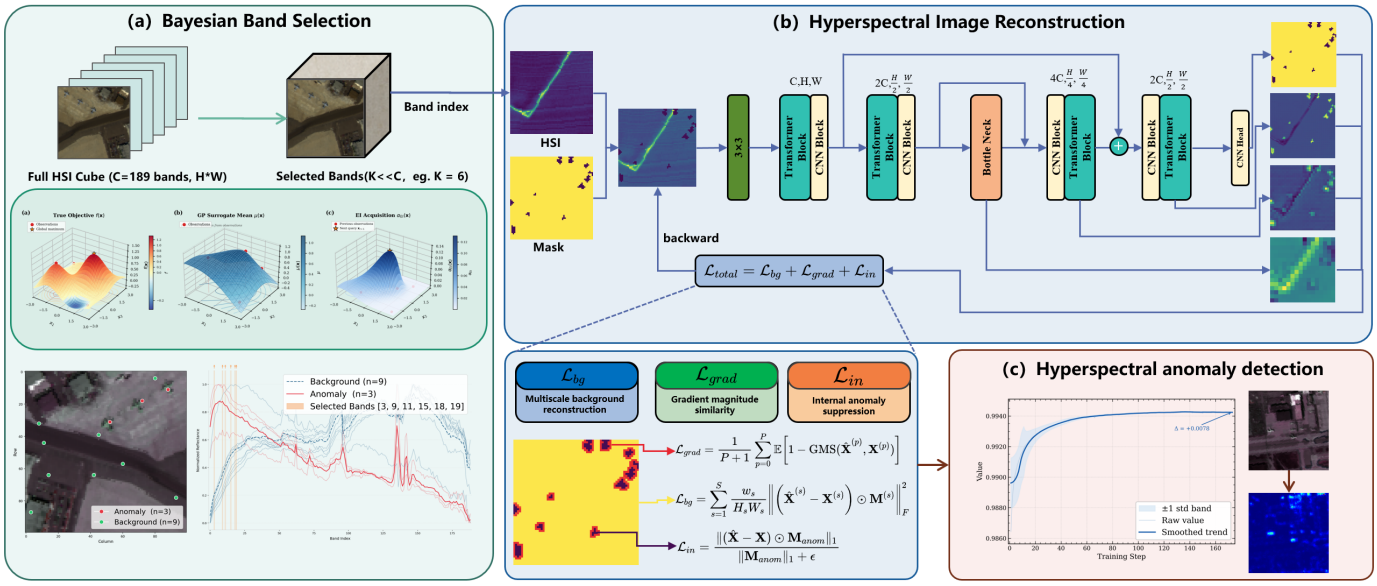


Fig. 2. Overall architecture of BBO-MARNet. (a) **Bayesian Band Selection** uses offline AUC feedback from a calibration/reference scene to select a task-aware spectral subset. (b) **Hyperspectral Image Reconstruction** performs mask-aware spatial-spectral reconstruction in the selected subspace, trained with the composite loss $\mathcal{L}_{total} = \mathcal{L}_{bg} + \mathcal{L}_{grad} + \mathcal{L}_{in}$. (c) **Hyperspectral Anomaly Detection** applies a statistical detector to the scoring tensor described in Section III-E to produce the anomaly score map. The band indices selected in (a) are fixed before target-scene reconstruction and detection in (b)–(c).

probability map $\mathbf{A} \in \mathbb{R}^{H\times W\times 1}$, effectively disentangling anomalous targets from complex background clutters.

B. Task-Aware Bayesian Band Selection

Hyperspectral images inherently suffer from severe spectral redundancy and inter-band collinearity. Processing the complete spectral data cube not only incurs prohibitive computational costs but also exacerbates the Hughes phenomenon, where the inclusion of uninformative or noisy bands actively degrades anomaly detection capabilities. Conventional band selection techniques predominantly rely on heuristic statistical metrics (e.g., variance or mutual information) that are entirely decoupled from the downstream task. To bridge this methodological gap, we formulate band selection as a task-aware combinatorial optimization problem, aiming to identify a minimal subset of bands that directly maximizes the end-to-end detection performance.

Mathematically, let the selected spectral subset be parameterized by a binary indicator vector $\mathbf{s} \in \{0,1\}^B$, where $s_b = 1$ denotes the inclusion of the b-th band from the original hyperspectral cube $\mathbf{X}$. The optimization is governed by a strict cardinality constraint $\|\mathbf{s}\|_0 = K$, where $K \ll B$ is the target number of retained bands. The objective is to determine the optimal configuration $\mathbf{s}^*$ that maximizes a task-specific performance metric $\mathcal{J}(\mathbf{s})$ (e.g., the Area Under the Curve (AUC) yielded by a lightweight surrogate detector such as the RX algorithm):

$$\mathbf{s}^* = \arg \max_{\mathbf{s}\in\{0,1\}^B, \|\mathbf{s}\|_0 = K} \mathcal{J}(\mathbf{s}). \tag{1}$$

Solving Eq. (1) via exhaustive search is computationally intractable, and the discrete objective function $\mathcal{J}(\mathbf{s})$ lacks an analytical gradient. Therefore, Bayesian Optimization (BO) is adopted to efficiently navigate this combinatorial search space. We explicitly distinguish this offline band-calibration step from the target-scene anomaly detection stage. When AUC is selected as $\mathcal{J}(\mathbf{s})$, the feedback necessarily requires GT on a calibration/reference scene; however, no GT from the evaluated target scene is used for band selection, network training, early stopping, threshold tuning, or inference. Thus, BBO-MARNet should be interpreted as an unsupervised target-scene detector with cross-scene band calibration, rather than as a fully label-free band-search algorithm. If no labeled calibration/reference scene is available, the AUC-driven BBO module can be disabled or replaced by an unsupervised proxy objective, but the reconstruction and detection stages remain label-free for the target scene. The exact dataset splitting and transfer protocols are systematically detailed in Section IV-B. To model the non-linear response surface across discrete band combinations, a Matérn 5/2 covariance kernel is employed. Notably, while the search space is discrete, the Euclidean distance d between two binary indicator vectors intrinsically correlates with their structural Hamming distance, enabling a valid continuous relaxation for the GP:

$$k(\mathbf{s},\mathbf{s}') = \sigma_f^2 \left(1 + \sqrt{5}\cdot d + \frac{5}{3}d^2\right)\exp(-\sqrt{5}\,d), \tag{2}$$

where $d = \frac{\|\mathbf{s}-\mathbf{s}'\|_2}{l}$ is the scaled Euclidean distance between the encoded combination vectors, l acts as the length-scale parameter, and σ_f^2 denotes the signal variance.

To balance exploration and exploitation, we use the Expected Improvement (EI) acquisition function. At BO iteration i, the GP is fitted to the observation history $\mathcal{D}_{i-1}$, yielding posterior mean $\mu_{i-1}(\mathbf{s})$ and standard deviation $\sigma_{i-1}(\mathbf{s})$. Throughout this finite search, $\mathbf{s}^*$ denotes the current best-observed subset in $\mathcal{D}_{i-1}$, an empirical approximation to the optimum defined in Eq. (1). Its observed performance $\mathcal{J}(\mathbf{s}^*)$ is

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the improvement baseline. The next candidate $\mathbf{s}^{(i)}$ is selected by maximizing

$$\mathrm{EI}(\mathbf{s} \mid \mathcal{D}_{i-1}) = \mathbb{E}[\max(0, \mathcal{J}(\mathbf{s}) - \mathcal{J}(\mathbf{s}^*)) \mid \mathcal{D}_{i-1}]. \quad (3)$$

For a Gaussian posterior, this acquisition function has the analytic form

$$\mathrm{EI}(\mathbf{s} \mid \mathcal{D}_{i-1}) = \begin{cases} \Delta_i(\mathbf{s})\,\Phi(Z_i) + \sigma_{i-1}(\mathbf{s})\,\phi(Z_i), & \sigma_{i-1}(\mathbf{s}) > 0, \\ 0, & \sigma_{i-1}(\mathbf{s}) = 0, \end{cases} \quad (4)$$

where $\Delta_i(\mathbf{s}) = \mu_{i-1}(\mathbf{s}) - \mathcal{J}(\mathbf{s}^*)$ and $Z_i = \Delta_i(\mathbf{s})/\sigma_{i-1}(\mathbf{s})$ for $\sigma_{i-1}(\mathbf{s}) > 0$. Here, $\Phi(\cdot)$ and $\phi(\cdot)$ are the standard normal cumulative distribution function and probability density function, respectively.

After N_{iter} BO iterations, the returned $\mathbf{s}^*$ is the best-observed subset in the final history $\mathcal{D}_{N_{\mathrm{iter}}}$. The selected bands form the sub-cube $\mathbf{X}_{\mathcal{S}} \in \mathbb{R}^{H\times W\times K}$ for subsequent mask-guided reconstruction. Algorithm 1 summarizes the procedure. We use b for a band index, j for an initial-sample index, and i for a BO iteration; the superscript (i) on $\mathbf{s}^{(i)}$ labels a candidate subset, not an individual band.

Algorithm 1 Task-Aware Bayesian Band Selection

Input: Calibration/reference-scene cube $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ and its GT for AUC feedback, target band count K, number of BO iterations N_{iter}, initial sample size N_{init}.
Output: Best-observed spectral subset indicator $\mathbf{s}^*$.

Initialization:
Define $\Omega = \{\mathbf{s} \in \{0,1\}^B : \|\mathbf{s}\|_0 = K\}$.
Sample N_{init} distinct initial configurations $\{\mathbf{s}_j^{\mathrm{init}}\}_{j=1}^{N_{\mathrm{init}}}$ from Ω.
for $j = 1$ to N_{init} **do**
  Extract sub-cube $\mathbf{X}_{\mathbf{s}_j^{\mathrm{init}}}$ and execute RX detection.
  Evaluate $y_j^{\mathrm{init}} \leftarrow \mathrm{AUC}_{\mathrm{cal}}(\mathbf{s}_j^{\mathrm{init}})$.
end for
Initialize $\mathcal{D}_0 \leftarrow \{(\mathbf{s}_j^{\mathrm{init}}, y_j^{\mathrm{init}})\}_{j=1}^{N_{\mathrm{init}}}$.
Bayesian Optimization Loop:
for $i = 1$ to N_{iter} **do**
  Fit the GP to $\mathcal{D}_{i-1}$ using the Matérn 5/2 kernel, obtaining μ_{i-1} and σ_{i-1}.
  Set $\mathbf{s}^* \leftarrow \arg\max_{\mathbf{s}}\{y : (\mathbf{s}, y) \in \mathcal{D}_{i-1}\}$.
  Uniformly sample a finite candidate pool $\mathcal{C}$ from unobserved configurations in Ω.
  Propose $\mathbf{s}^{(i)} \leftarrow \arg\max_{\mathbf{s}\in\mathcal{C}} \mathrm{EI}(\mathbf{s} \mid \mathcal{D}_{i-1})$.
  Extract $\mathbf{X}_{\mathbf{s}^{(i)}}$ and execute RX detection.
  Evaluate $y^{(i)} \leftarrow \mathrm{AUC}_{\mathrm{cal}}(\mathbf{s}^{(i)})$.
  Update $\mathcal{D}_i \leftarrow \mathcal{D}_{i-1} \cup \{(\mathbf{s}^{(i)}, y^{(i)})\}$.
end for
Return $\mathbf{s}^* \leftarrow \arg\max_{\mathbf{s}}\{y : (\mathbf{s}, y) \in \mathcal{D}_{N_{\mathrm{iter}}}\}$.

C. Mask-Guided Spatial-Spectral Network

Conventional deep autoencoders employed in hyperspectral anomaly detection frequently suffer from the "identity mapping" dilemma: owing to their excessive representational capacity, they tend to trivially memorize the pixel intensities of the input cube, thereby inadvertently reconstructing both the background and the anomalous targets. To circumvent this issue and compel the network to capture the intrinsic background manifold, a mask-guided self-supervised reconstruction mechanism is introduced. Furthermore, to address the distinct physical characteristics of the spatial and spectral dimensions in hyperspectral imagery, a dual-branch U-shaped spatial-spectral Transformer architecture is explicitly designed.

Let the band-optimized hyperspectral cube be denoted as $\mathbf{X}_{\mathcal{S}} \in \mathbb{R}^{H\times W\times K}$. Prior to feature extraction, a binary spatial masking operator $\mathcal{M}(\cdot)$ is applied to intentionally occlude localized spatial patches. The masked input cube can be formulated as:

$$\mathbf{X}_{in} = \mathbf{X}_{\mathcal{S}} \odot \mathbf{M}_{mask}, \quad (5)$$

where $\mathbf{M}_{mask} \in \{0,1\}^{H\times W\times 1}$ is the binary mask tensor (with 0 indicating occluded regions and 1 indicating visible regions), and $\odot$ denotes element-wise multiplication broadcasted along the spectral dimension. To physically simulate the random distribution and varying scales of real-world anomalies, the mask generation process is fundamentally stochastic. Specifically, during each forward pass, the total number of occluded patches (representing synthetic anomalous targets), denoted as N, is randomly drawn from a discrete uniform distribution $N \sim \mathcal{U}(1, 32)$. Concurrently, the spatial footprint A of each individual patch is uniformly sampled from $A \sim \mathcal{U}(3, 20)$ pixels. By completely corrupting the spectral signatures within these stochastically masked regions, the network establishes a conditional reconstruction dynamic. Operating strictly within the previously optimized material-sensitive subspace, the architecture is forced to hallucinate the missing content utilizing long-range contextual priors from the unmasked background. This mechanism fundamentally deprives the network of the spectral shortcuts required to reconstruct rare target materials, explicitly amplifying the residual discrepancy between the anomalous targets and the background manifold, rather than merely preventing identity mapping.

The corrupted cube $\mathbf{X}_{in}$ is first mapped to a shallow feature space $\mathbf{F}_0 \in \mathbb{R}^{H\times W\times C}$ through a 3×3 convolution, where C denotes the base feature dimension. The resulting feature is then processed by a hierarchical U-shaped encoder-decoder. At each level, feature transformation is performed by the Spatial-Spectral Transformer Block illustrated in Fig. 3. The block contains two sequential and complementary stages rather than two parallel branches. The spectral Transformer stage first models global inter-channel correlations and supplements them with local spatial encoding. Its output is subsequently processed by the spatial Transformer stage, in which W-MSA and SW-MSA capture within-window and cross-window dependencies, respectively.

1) Spectral Transformer Branch: To capture the complex inter-band correlations and global spectral similarities without incurring the $\mathcal{O}((HW)^2)$ computational bottleneck of standard spatial attention, a transposed cross-covariance attention mechanism is implemented. Given an input feature map $\mathbf{F}^{(l-1)}$, it is linearly projected into Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value

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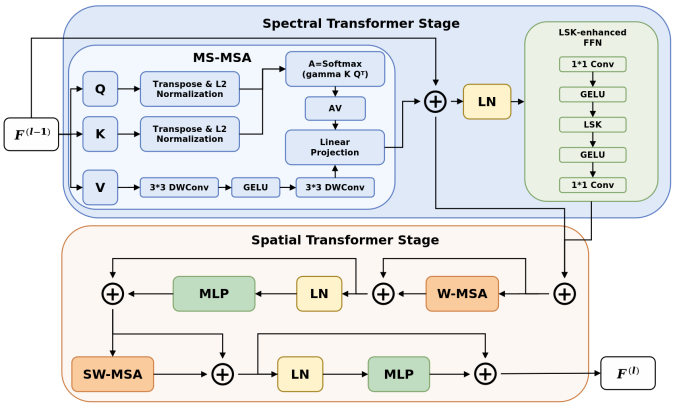


Fig. 3. Detailed architecture of the Spatial-Spectral Transformer Block. The spectral Transformer stage first performs transposed channel-covariance attention through MS-MSA and complements it with a local depth-wise convolutional path, followed by an LSK-enhanced feed-forward sublayer. The resulting representation is then processed sequentially by W-MSA and SW-MSA blocks to model within-window and cross-window spatial dependencies, respectively. Each sublayer is equipped with the residual connection used in the implementation.

($\mathbf{V}$) tensors. The spectral attention map $\mathbf{A}_{spec} \in \mathbb{R}^{C\times C}$ is computed across the channel dimension:

$$\mathbf{A}_{spec} = \mathrm{Softmax}\Big(\boldsymbol{\gamma} \cdot \widetilde{\mathbf{K}}\widetilde{\mathbf{Q}}^{\mathsf{T}}\Big), \tag{6}$$

where $\widetilde{\mathbf{K}} = \mathrm{Norm}_2(\mathbf{K})$ and $\widetilde{\mathbf{Q}} = \mathrm{Norm}_2(\mathbf{Q})$. With $\mathbf{K}, \mathbf{Q} \in \mathbb{R}^{C\times(HW)}$, Norm_2 applies ℓ_2-normalization to each channel vector along the flattened spatial dimension. Thus, $\widetilde{\mathbf{K}}\widetilde{\mathbf{Q}}^{\mathsf{T}} \in \mathbb{R}^{C\times C}$, and the transpose acts only on the normalized query matrix. The learnable scaling parameter $\boldsymbol{\gamma}$ is retained. The spectral global representation is then fused with a parallel local spatial convolution branch to prevent the loss of high-frequency structural details during the channel interaction:

$$\hat{\mathbf{F}}_{spec} = \mathbf{A}_{spec}\mathbf{V} + \mathrm{DWConv}_{3\times3}(\mathbf{V}) + \mathbf{F}^{(l-1)}, \tag{7}$$

where $\mathrm{DWConv}_{3\times3}(\cdot)$ denotes a depth-wise convolution acting as a positional encoding complement.

2) Spatial Transformer Branch: Following the global spectral calibration, the features are processed by a shifted-window spatial Transformer (Swin) module to explicitly model local structural textures and cross-window boundaries. It computes standard multi-head self-attention (MSA) within non-overlapping local windows (W-MSA), followed by a shifted-window configuration (SW-MSA) to enable cross-window spatial interactions:

$$\hat{\mathbf{F}}_{spa} = \text{W-MSA}(\mathrm{LN}(\hat{\mathbf{F}}_{spec})) + \hat{\mathbf{F}}_{spec}, \tag{8}$$

$$\mathbf{F}^{(l)} = \text{SW-MSA}(\mathrm{LN}(\hat{\mathbf{F}}_{spa})) + \hat{\mathbf{F}}_{spa}, \tag{9}$$

where $\mathrm{LN}(\cdot)$ denotes Layer Normalization. This sequential design ensures that the network characterizes the hyperspectral data comprehensively, capturing both continuous spectral manifolds and structured spatial topographies.

In the decoder phase, the deep abstract features are progressively upsampled and concatenated with the high-resolution features preserved from the corresponding encoder stages via skip connections. This multiscale feature fusion mitigates the spatial degradation caused by downsampling operations. Ultimately, a convolutional projection head maps the decoded features back to the original spectral dimension, supplemented by a global residual connection to yield the final reconstructed cube $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times K}$:

$$\hat{\mathbf{X}} = \mathrm{Conv}(\mathbf{F}_{out}) + \mathbf{X}_{in}. \tag{10}$$

By predicting the missing regions exclusively from the surrounding context, the network establishes a robust background representation. The discrepancy between the reconstructed cube $\hat{\mathbf{X}}$ and the original spectral subset $\mathbf{X}_{\mathcal{S}}$—particularly inside the masked boundaries—serves as a critical indicator of anomaly presence, which systematically guides the design of the composite loss function detailed in the subsequent section.

D. Composite Loss Function Optimization

To effectively optimize the proposed mask-guided spatial-spectral network and guarantee the disentanglement of the background manifold from anomalous targets, a tailored composite objective function is formulated. The total loss function $\mathcal{L}_{total}$ comprises three complementary constraints: a multi-scale masked background reconstruction loss ($\mathcal{L}_{bg}$), a multi-scale gradient magnitude similarity loss ($\mathcal{L}_{grad}$), and an internal robust regularization loss ($\mathcal{L}_{in}$). The joint optimization objective is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{bg} + \mathcal{L}_{grad} + \mathcal{L}_{in}. \tag{11}$$

1) Multi-Scale Masked Background Reconstruction Loss ($\mathcal{L}_{bg}$): To capture hierarchical contextual semantics and ensure global spectral fidelity, the background reconstruction is explicitly supervised across multiple spatial scales. Let S denote the total number of hierarchical scales extracted from the decoder ($S = 3$, corresponding to spatial resolutions of $1/4$, $1/2$, and 1). The masked reconstruction error is evaluated strictly on the visible background regions:

$$\mathcal{L}_{bg} = \sum_{s=1}^{S} \frac{w_s}{H_s\, W_s} \left\| \left(\hat{\mathbf{X}}^{(s)} - \mathbf{X}^{(s)}\right) \odot \mathbf{M}^{(s)} \right\|_F^2, \tag{12}$$

where $\mathbf{X}^{(s)}$, $\hat{\mathbf{X}}^{(s)}$, and $\mathbf{M}^{(s)}$ represent the downsampled original cube, the reconstructed feature map, and the background mask at scale s, respectively. The balancing weights are empirically assigned as $w_s \in \{0.25, 0.5, 1.0\}$ to allocate ascending focus toward the highest-resolution output. This multi-level supervision enforces geometric consistency and alleviates the optimization difficulty of deep representations.

2) Multi-Scale Gradient Magnitude Similarity Loss ($\mathcal{L}_{grad}$): Standard pixel-wise regression losses equally penalize all pixels, frequently resulting in over-smoothed spatial textures and blurred object boundaries. To mitigate this phenomenon, a Gradient Magnitude Similarity (GMS) constraint is introduced to preserve high-frequency contour continuity. Utilizing a multidirectional Sobel operator ∇, the structural similarity between the reconstruction and the original input is quantified:

$$\mathrm{GMS}(\hat{\mathbf{X}}, \mathbf{X}) = \frac{2\|\nabla\hat{\mathbf{X}}\|_2 \odot \|\nabla\mathbf{X}\|_2 + c}{\|\nabla\hat{\mathbf{X}}\|_2^2 + \|\nabla\mathbf{X}\|_2^2 + c}, \tag{13}$$

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where c is a small constant ensuring numerical stability. To acquire robust structural awareness regardless of target scale, this metric is evaluated across a P-level average-pooling pyramid:

$$\mathcal{L}_{grad} = \frac{1}{P+1}\sum_{p=0}^{P} \text{Mean}\left(1 - \text{GMS}(\hat{\mathbf{X}}^{(p)}, \mathbf{X}^{(p)})\right). \quad (14)$$

Geometrically, this term guarantees that the inferred background smoothly aligns with the surrounding topological structures, thereby reducing false alarms triggered by naturally complex boundaries.

3) Internal Robust Regularization Loss ($\mathcal{L}_{in}$): To regulate the generative behavior strictly within the occluded regions ($\mathbf{M}_{anom} = 1 - \mathbf{M}$) without inadvertently forcing the network to memorize anomalous targets, an internal ℓ_1-norm regularization is applied:

$$\mathcal{L}_{in} = \frac{\|(\hat{\mathbf{X}} - \mathbf{X}) \odot \mathbf{M}_{anom}\|_1}{\|\mathbf{M}_{anom}\|_1 + \epsilon}. \quad (15)$$

Unlike the ℓ_2-norm, which heavily penalizes large errors and forces the model to fit outliers, the ℓ_1-norm provides a robust statistical constraint that is inherently tolerant to sparse outliers. This crucial property prevents the large error gradients generated by actual anomalies within the mask from dominating the optimization process. Consequently, the network is encouraged to hallucinate the missing content based on prevalent global background priors rather than precisely fitting local anomalous irregularities, thereby preserving a distinct residual gap for subsequent detection.

Optimization Strategy: The entire end-to-end framework is optimized using the Adaptive Moment Estimation (Adam) optimizer. The initial learning rate is set to 1×10^{-4}. To appropriately constrain the parameter space and prevent overfitting—particularly given the limited sample sizes characteristic of hyperspectral datasets—a strict weight decay (L2 penalty) of 5×10^{-6} is applied. The model is trained over 400 epochs with a mini-batch size of 16, ensuring stable convergence and robust cross-scene generalization.

E. Anomaly Scoring and Detection Stage

After training, the network produces a reconstructed background cube $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times K}$ for the band-selected input $\mathbf{X}_{\mathcal{S}}$. The final detector is not trained with GT labels. Instead, an anomaly score is computed by applying a statistical detector to a selected scoring tensor. In the default BBO-MARNet setting, the normalized input and reconstruction are fused as

$$\mathbf{Z} = \text{Norm}(\mathbf{X}_{\mathcal{S}}) + \text{Norm}(\hat{\mathbf{X}}), \quad (16)$$

where $\text{Norm}(\cdot)$ denotes min-max normalization within the selected spectral subspace. For each pixel vector $\mathbf{z}_i \in \mathbb{R}^K$, the RX score is then calculated as

$$a_i = (\mathbf{z}_i - \boldsymbol{\mu})^T(\boldsymbol{\Sigma} + \epsilon\mathbf{I})^{-1}(\mathbf{z}_i - \boldsymbol{\mu}), \quad (17)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are estimated from all pixels in $\mathbf{Z}$ and ϵ is a small diagonal stabilizer. The resulting score map $\mathbf{A} = \{a_i\}$ is thresholded only for evaluation or visualization. This design differs from a pure reconstruction-error detector: the reconstruction error $|\mathbf{X}_{\mathcal{S}} - \hat{\mathbf{X}}|$ is evaluated as the *Dif* protocol in the ablation study, whereas the final detector uses the fused representation because it empirically preserves discriminative target spectra while benefiting from background purification.

IV. Experiments

To validate the effectiveness, efficiency, and generalization capability of the proposed method, comprehensive experiments were conducted on five hyperspectral datasets, including four public benchmarks and one newly curated maritime anomaly dataset. The experimental protocol, implementation details, and evaluation metrics are described below, followed by quantitative and qualitative results.

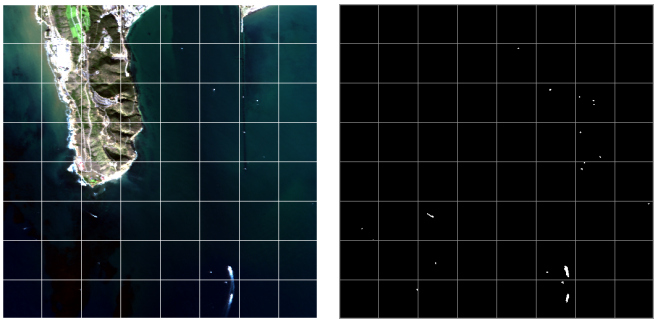
A. Datasets

Five datasets (spanning seven distinct evaluation scenes) were utilized to evaluate the proposed framework. To ensure a comprehensive assessment, these include both conventional single-scene benchmarks and large-scale patch-based datasets:

- **HAD100 Dataset [25]:** This is a large-scale hyperspectral anomaly detection benchmark containing 100 real test scenes acquired by AVIRIS. The scenes are uniformly cropped into 64×64 pixel patches, encompassing up to 276 spectral bands. Diverse small targets, such as vehicles and buildings, are labeled as anomalies, typically occupying from a few to tens of pixels per scene.
- **San Diego Dataset [13]:** Collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the San Diego naval airport, CA, USA. The widely adopted sub-scene has spatial dimensions of 100×100 pixels. After discarding bands affected by water vapor absorption and low signal-to-noise ratios, 189 spectral bands are retained. Within this area, three airplanes are classified as anomalies, occupying 57 pixels in total.
- **ABU Dataset [37]:** The Airport-Beach-Urban (ABU) dataset is a comprehensive benchmark collection comprising 13 different scenes captured primarily by the AVIRIS sensor. In our experiments, three representative sub-scenes (denoted as ABU-a, ABU-b, and ABU-u) with varying spatial dimensions and spectral bands were selected. In these complex backgrounds, specific targets like localized vehicles, airplanes, and structures are manually labeled as anomalies.
- **Texas Coast Dataset [37]:** This dataset originates from an urban coastal area monitored by the AVIRIS sensor. The selected scene has dimensions of 100×100 pixels with a spatial resolution of 17.2 m per pixel. After preprocessing, 204 bands are utilized. Against this natural coastal background, specific man-made buildings are considered anomalies, occupying 67 pixels.
- **Ship Dataset (Ours):** This newly constructed benchmark is specifically tailored for hyperspectral anomaly detection in complex maritime and coastal environments and was acquired by AVIRIS. The panoramic scene has spatial dimensions of 512×512 pixels with a ground sample distance of 4.0 m and comprises 224 spectral

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(a) Pseudo-color image (b) Ground-truth map

Fig. 4. The Ship dataset for hyperspectral anomaly detection. (a) Pseudo-color image. (b) Binary ground-truth map with the non-overlapping 64×64 grid used for dataset splitting. Patches without annotated ship pixels are used for training, whereas patches containing at least one annotated ship pixel are reserved for testing.

bands spanning the visible to shortwave infrared range (approximately 365.9-2496.2 nm). As illustrated by the pseudo-color composite in Fig. 4a, the scene contains highly heterogeneous backgrounds consisting of both terrestrial landmasses and extensive sea surfaces. The ship ground truth, comprising 231 annotated pixels, was manually delineated by a single expert using ENVI following a consistent annotation protocol, thereby avoiding inter-annotator variability. Candidate ship regions were first identified from false-color composites and were subsequently refined through pixel-wise inspection and spectral comparison with the surrounding background. The finalized annotations were converted into a binary ground-truth mask, with ship pixels labeled as anomalies and all remaining pixels as background. The 512×512 scene was systematically partitioned into 64 non-overlapping 64×64 patches for training/testing, as shown in Fig. 4b. A patch was assigned to the training set only when it contained no annotated ship pixels; any patch containing at least one ship pixel was reserved for testing. This criterion was adopted to ensure that the reconstruction network was trained exclusively on target-free background patches while all target-containing patches were retained for evaluation. The resulting split contains 52 background-only training patches and 12 target-containing testing patches, with no patch- or pixel-level overlap between the two sets.

Figure 5 presents the pseudo-color composites and corresponding ground-truth maps of the five investigated datasets, covering all seven evaluation scenes used in our experiments. For HAD100, one representative test scene is displayed for conciseness, while all 100 test scenes are retained in the quantitative evaluation.

Regarding the experimental setup, the HAD100 and Ship datasets operate under a split-set paradigm, utilizing designated pure background samples for representation learning and separate patches for evaluation. Conversely, for the conventional single-scene datasets (San Diego, Texas Coast, and the ABU sub-scenes), the entire image is utilized directly for background modeling without prior target exclusion. In these cases, the ground truth is strictly reserved for the final evaluation phase to ensure a rigorous anomaly detection protocol.

TABLE I
SUMMARY OF THE INVESTIGATED HYPERSPECTRAL ANOMALY DETECTION DATASETS

Dataset	Size ($h \times w \times c$)	S.R.	Anomaly Type	Ratio (%)
Ship	$64 \times 64 \times 224$	4.0	Ship	$0.47^{\dagger}$
HAD100	$64 \times 64 \times 276$	$\leq$8.4	Varies	$0.87^{\dagger}$
San Diego	$100 \times 100 \times 189$	3.5	Plane	0.57
ABU-a	$100 \times 100 \times 205$	7.1	Plane	1.44
ABU-b	$150 \times 150 \times 188$	17.2	Ship	0.08
ABU-u	$100 \times 100 \times 191$	3.5	Building	0.52
Texas Coast	$100 \times 100 \times 204$	17.2	Building	0.67

**S.R.: Spatial Resolution (m).*

†For datasets with a train/test split (i.e., Ship and HAD100), the anomaly ratio is calculated based solely on the testing patches.

B. Implementation Details

The BBO-MARNet framework is implemented using PyTorch on dual NVIDIA GeForce RTX 3090 Ti GPUs. For the Bayesian Band Optimization (BO) phase, we strictly implemented a cross-scene evaluation protocol to ensure experimental fairness and verify the transferability of the selected bands:

- **Large-scale Cross-Calibration:** For datasets with diverse patches, we adopted a reciprocal selection strategy. Specifically, the optimal band subset for the HAD100 dataset was determined by performing BO on the Ship dataset, and vice versa. This mutual calibration ensures that the spectral prioritization is robust to large-scale scene variations.
- **Single-Scene Transfer:** For individual benchmark scenes, the BO process utilized a reference scene containing structurally similar anomalous targets. For example, when evaluating the ABU-a dataset (containing aircraft targets), the BO module was first executed on the San Diego dataset using its ground truth to identify the detection-optimal bands. These bands were then frozen and utilized for training and testing on the ABU-a dataset without any further access to ABU-a's ground truth during the optimization or training phases.

Under both protocols, the selected band indices are frozen before the target-scene reconstruction network is trained. Target-scene ground truth is never used to update BO observations, train the reconstruction network, perform early stopping, select detection thresholds, or tune hyperparameters.

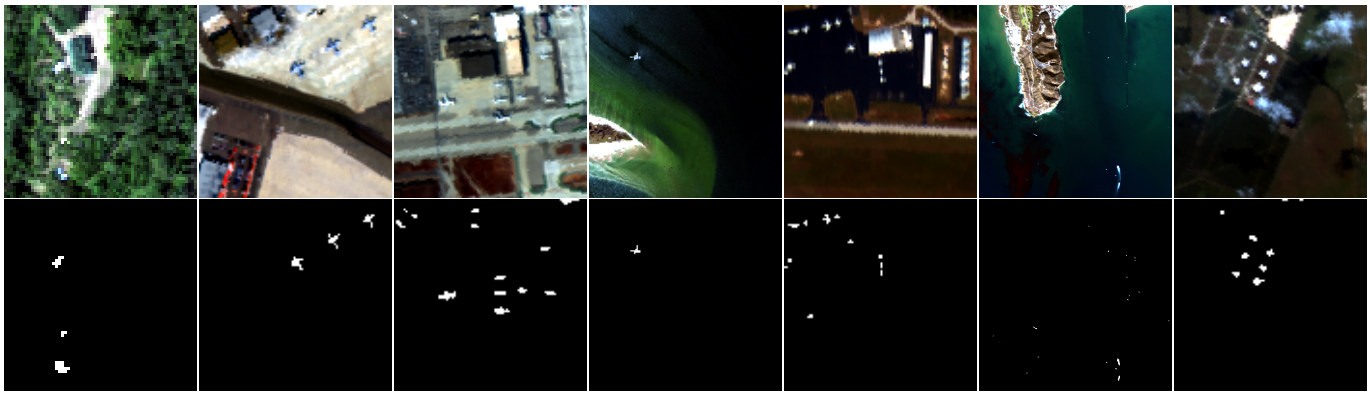
The reconstruction network was trained for 400 epochs using the Adam optimizer with a learning rate of 1×10^{-4} and a weight decay of 5×10^{-6}. An early stopping patience of 40 epochs was applied to ensure stable convergence.

C. Evaluation Metrics

To quantitatively assess the anomaly detection performance, the receiver operating characteristic (ROC) curve and the area under the curve (AUC) serve as the primary evaluation metrics. The ROC curve delineates the dynamic trade-off between the probability of detection (P_D) and the probability of false alarm (P_F) across varying segmentation thresholds.

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(a) HAD100 (b) San Diego (c) ABU-a (d) ABU-b (e) ABU-u (f) Ship (g) Texas Coast

Fig. 5. Pseudo-color composites and the corresponding ground-truth maps of the five investigated datasets, covering the seven evaluation scenes used in our experiments: (a) a representative HAD100 test scene, (b) San Diego, (c) ABU-a, (d) ABU-b, (e) ABU-u, (f) Ship, and (g) Texas Coast. White pixels in the ground-truth maps denote anomalies.

The AUC yields a comprehensive scalar summarizing the overall detection capability by integrating the ROC curve, mathematically defined as:

$$\mathrm{AUC} = \int_0^1 P_D(P_F)\, dP_F, \tag{18}$$

where a higher AUC value consistently indicates superior detection robustness.

To accommodate the structural diversity of the evaluated datasets, the evaluation protocol is adapted accordingly. For the large-scale patch-based datasets (i.e., HAD100 and the proposed Ship dataset), the Mean AUC (mAUC) is adopted. Specifically, the standard AUC is computed independently for each test patch, and the global mAUC is obtained by averaging these patch-wise scores:

$$\mathrm{mAUC} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{AUC}_i, \tag{19}$$

where N denotes the total number of test patches, and AUC_i represents the score of the i-th patch. Conversely, for conventional single-scene hyperspectral benchmarks, the standard global AUC metric is directly applied.

In addition to the full-range area under the ROC curve (AUC), we report average precision (AP), which summarizes the precision-recall curve and is more informative under severe class imbalance. The maximum F1 score is computed over all thresholds as

$$F1_{\max} = \max_{\tau} \frac{2P(\tau)R(\tau)}{P(\tau)+R(\tau)}. \tag{20}$$

To evaluate the operationally important low-false-alarm region, we select

$$\tau^* = \arg\max_{\tau:\, P_F(\tau)\le 0.01} R(\tau), \tag{21}$$

and report $P_{1\%} = P(\tau^*)$ and $R_{1\%} = R(\tau^*)$. We also report the normalized partial AUC

$$\mathrm{pAUC}_{1\%} = \frac{1}{0.01}\int_0^{0.01} P_D(P_F)\, \mathrm{d}P_F. \tag{22}$$

All threshold-dependent quantities are used only for post-hoc evaluation and not for training or model selection.

Furthermore, to comprehensively evaluate the operational viability and computational efficiency of the models, the inference time (measured in milliseconds per image) and model complexity (total learnable parameters) are systematically recorded.

D. Comparison with State-of-the-Art Methods

1) Quantitative Analysis: The quantitative detection performance of the proposed methods and comparative baselines across the seven diverse datasets is comprehensively reported in Table II. To facilitate comparison, the best and second-best results for each dataset are indicated by boldface and an underline, respectively. The evaluated methods are broadly categorized into deep learning-based models, traditional statistical detectors, and our proposed frameworks (**MARNet** and **BBO-MARNet**).

Traditional statistical and machine learning approaches, including RX, LRX, and SVM, exhibit severe performance fluctuations across different scenes. For instance, while LRX achieves a competitive score of 0.99367 on the Ship dataset, its performance experiences a precipitous decline to 0.73359 and 0.90880 on the San Diego and HAD100 datasets, respectively. This instability primarily stems from their reliance on rigid statistical assumptions (e.g., multivariate Gaussian distribution for the background), which inevitably fail to characterize the complex, highly heterogeneous background clutter present in real-world hyperspectral imagery.

Deep learning-based models generally demonstrate superior background suppression capabilities compared to traditional detectors. Methods such as AETNet, FERD, and DFAN maintain relatively high and stable performance across most datasets, benefiting from the robust feature extraction inherent in deep neural networks. Nevertheless, several advanced networks still struggle with cross-scene generalization. Specifically, BSRegNet and UADNet experience significant performance drops on the challenging ABU-a dataset (0.73399 and 0.85341, respectively), indicating that these models are

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TABLE II
PERFORMANCE COMPARISON OF VARIOUS MODELS ON DIFFERENT DATASETS

model/dataset	HAD100	San Diego	ABU-a	ABU-b	ABU-u	Ship	Coast
Auto-AD [38]	-	0.98229	0.94744	0.99261	0.99076	-	0.99287
BSRegNet [23]	-	0.90407	0.73399	0.95247	0.99256	-	0.98865
MSNet [39]	-	0.99154	0.94442	0.97898	0.98094	-	0.99722
DirectNet [20]	-	0.98708	0.91964	0.98128	0.97767	-	0.99053
AETNet [25]	0.99254	0.97995	0.94203	0.99600	0.97197	0.99755	0.99518
FERD [40]	0.99321	0.96588	0.94968	0.99443	0.98210	0.99875	0.99641
BSDM [6]	0.98513	0.97525	0.90282	0.98790	0.97161	0.99943	0.99217
ERX [41]	0.91462	0.98573	0.88024	0.92104	0.98233	0.99510	0.97485
UADNet [42]	0.87115	0.98468	0.85341	0.95401	0.96182	0.99892	0.98865
DFAN [43]	0.98314	0.97552	0.95037	0.99719	0.97491	0.99956	0.99552
RX	0.96339	0.87363	0.82210	0.98075	0.95128	0.99858	0.99065
LRX	0.90880	0.73359	0.91494	0.99325	0.94377	0.99367	0.97159
SVM	0.93725	0.97654	0.69899	0.90626	0.88234	0.99922	0.96815
KMeans	0.94959	0.98157	0.84239	0.98456	0.93275	0.96804	0.95990
AE	0.95574	0.82200	0.86430	0.98696	0.90517	0.99755	0.93141
SLRX	0.91640	0.74621	0.92852	0.98320	0.97386	0.97372	0.98906
MARNet	**0.99431**	0.98890	0.97628	0.99727	**0.99343**	0.99930	0.99704
BBO-MARNet	0.99429	**0.99392**	**0.97671**	**0.99910**	0.99033	**0.99961**	**0.99791**

prone to overfitting specific background structures and struggle to separate weak anomalies from intricate local clutters.

In contrast, the proposed frameworks show competitive quantitative results across the evaluated datasets. **MARNet**, which sequentially employs representation reconstruction and a subsequent detector head, achieves the highest scores on the HAD100 (0.99431) and ABU-u (0.99343) datasets. This demonstrates that the proposed mask-guided reconstruction strategy effectively purifies the background representations, enabling the detector head to robustly delineate anomalous targets. Furthermore, by incorporating an initial task-aware band selection module prior to reconstruction, **BBO-MARNet** elevates the detection accuracy even further, achieving state-of-the-art performance on the San Diego, ABU-a, ABU-b, Ship, and Coast datasets. This quantitative improvement verifies that strategic Bayesian band optimization effectively mitigates the Hughes phenomenon and eliminates redundant spectral correlations. By feeding only the most informative spectral subset into the reconstruction network, **BBO-MARNet** minimizes the risk of background information leakage, thereby maximizing target saliency.

Beyond detection accuracy, we evaluate the practical deployability of the proposed framework in resource-constrained remote sensing environments by analyzing computational complexity, model size, and inference speed. BBO-MARNet is remarkably compact, comprising merely 1.69 M parameters and requiring 6.95 GFLOPs. This architectural footprint is nearly two orders of magnitude smaller than heavily parameterized networks such as FERD, which demands a substantial 158.99 M parameters and 60.02 GFLOPs. While ultra-lightweight architectures like MSNet (0.09 M parameters) and BSDM (1.08 M parameters) exhibit lower theoretical computational costs and exceptionally high frame rates, this extreme compression comes at a severe cost to cross-scene robustness, as evidenced by their degraded accuracy on the highly heterogeneous ABU-a dataset. Furthermore, in terms of practical execution speed evaluated on the San Diego dataset, the complete BBO-MARNet pipeline maintains a rapid inference speed of 82.70 FPS, comfortably exceeding the standard 30 FPS requirement for real-time dynamic remote sensing streams. In summary, rather than pursuing extreme but fragile network compression, BBO-MARNet establishes an optimal trade-off, consistently delivering state-of-the-art detection accuracy across diverse scenes without imposing prohibitive memory or computational burdens.

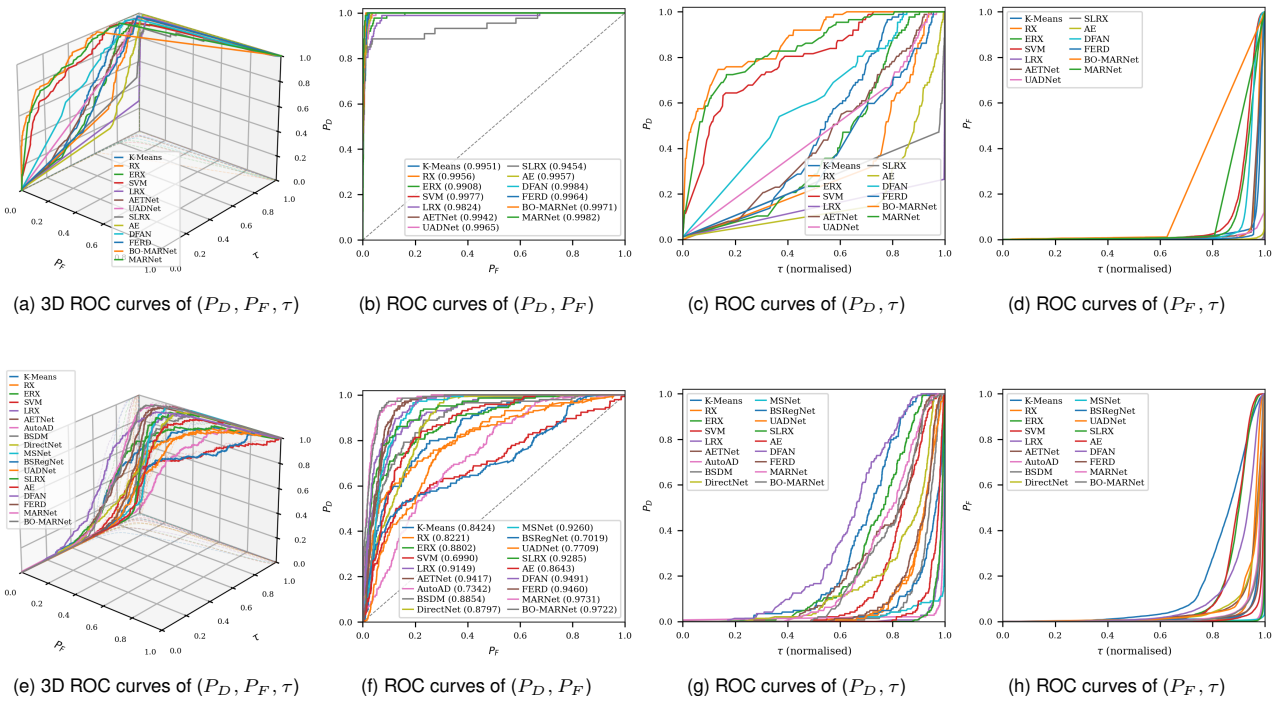
To further evaluate the target-background separability and threshold robustness of the proposed framework, comprehensive 3D ROC curves (P_D, P_F, τ) and their corresponding 2D projections are plotted for the Ship-53 and ABU-a datasets, as illustrated in Fig. 6. Unlike the traditional 2D ROC curve, the 3D ROC surface provides a holistic evaluation of the dynamic relationship between the probability of detection (P_D), the probability of false alarm (P_F), and the shifting segmentation threshold (τ).

As observed in the standard 2D ROC curves of (P_D, P_F) (Fig. 6(b) and (f)), the curves of the proposed MARNet and BBO-MARNet consistently wrap above those of all comparative baselines, approaching the top-left corner most rapidly. This steep initial ascent indicates that our methods achieve an exceptionally high detection rate even when the false alarm penalty is stringently constrained. Particularly on the challenging ABU-a dataset, which contains severe structured urban background interference, traditional statistical models and baseline reconstruction networks struggle to separate anomalies from background edges. In contrast, BBO-MARNet yields a highly competitive AUC, verifying its exceptional background suppression capability.

Furthermore, the (P_D, τ) and (P_F, τ) projection curves demonstrate the superior threshold stability of the proposed approach. An optimal hyperspectral anomaly detector should ideally maintain a high P_D and a near-zero P_F across a broad spectrum of threshold variations. As shown in Fig. 6(c), (d),

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(a) 3D ROC curves of (P_D, P_F, τ) (b) ROC curves of (P_D, P_F) (c) ROC curves of (P_D, τ) (d) ROC curves of (P_F, τ)

(e) 3D ROC curves of (P_D, P_F, τ) (f) ROC curves of (P_D, P_F) (g) ROC curves of (P_D, τ) (h) ROC curves of (P_F, τ)

Fig. 6. ROC curves evaluation on the Ship-53 and ABU-a datasets. (a) 3D ROC curves of (P_D, P_F, τ) for Ship-53. (b) 2D ROC curves of (P_D, P_F) for Ship-53. (c) 2D ROC curves of (P_D, τ) for Ship-53. (d) 2D ROC curves of (P_F, τ) for Ship-53. (e) 3D ROC curves of (P_D, P_F, τ) for ABU-a. (f) 2D ROC curves of (P_D, P_F) for ABU-a. (g) 2D ROC curves of (P_D, τ) for ABU-a. (h) 2D ROC curves of (P_F, τ) for ABU-a.

(g), and (h), BBO-MARNet and MARNet exhibit a remarkably slow decay in P_D as τ increases, while their P_F drops sharply to near-zero at very low threshold values. This substantial threshold tolerance signifies that the mask-aware spatial-spectral reconstruction dynamically enlarges the residual gap between anomalous targets and the background manifold, effectively preventing the representation degradation of anomalies.

Consequently, the 3D ROC surfaces of the proposed methods encapsulate the largest volume under the curve. This quantitative superiority firmly establishes an optimal and robust decision boundary, validating the operational reliability of BBO-MARNet for target recognition in complex remote sensing scenarios.

A high detection threshold may trivially suppress both background responses and true anomalies. Therefore, we evaluate background suppression at matched target sensitivities and report the corresponding (τ, P_F) pairs. For each target detection probability p, the operating threshold is selected as

$$\tau^*(p) = \max\{\tau \mid P_D(\tau) \geq p\},$$

which identifies the highest threshold that maintains the required detection sensitivity. All detection maps are normalized to $[0, 1]$ before thresholding to place the anomaly scores of different methods on a common numerical range. The ABU-a scene is selected for this quantitative analysis because valid archived detection maps are available for all 18 compared methods, enabling a consistent and comprehensive operating-point comparison.

Table III quantitatively complements the (P_F, τ) projections in Fig. 6 by comparing all 18 methods with available detection maps on the ABU-a scene. MARNet and BBO-MARNet achieve the two lowest false-alarm probabilities at all three sensitivity levels. BBO-MARNet ranks second at $P_D \geq 0.80$ and first at $P_D \geq 0.90$ and 0.95. Compared with the strongest non-proposed method at each operating point, BBO-MARNet reduces P_F by 40.37%, 47.91%, and 39.28%, respectively. At the two higher-sensitivity operating points,

TABLE III
QUANTITATIVE BACKGROUND-SUPPRESSION PERFORMANCE ON THE ABU-A SCENE. EACH ENTRY REPORTS τ/P_F UNDER A FIXED MINIMUM P_D. LOWER P_F INDICATES BETTER BACKGROUND SUPPRESSION; THE BEST AND SECOND-BEST VALUES ARE BOLDFACED AND UNDERLINED, RESPECTIVELY.

Method	$P_D \geq 0.80$ (τ/P_F)	$P_D \geq 0.90$ (τ/P_F)	$P_D \geq 0.95$ (τ/P_F)
Auto-AD	0.00207/0.473011	0.00139/0.583705	0.00110/0.643770
BSRegNet	0.00892/0.651989	0.00676/0.767451	0.00650/0.781757
MSNet	0.00031/0.115361	0.00023/0.164367	0.00021/0.191254
DirectNet	0.05813/0.212256	0.05011/0.253754	0.04538/0.287135
AETNet	0.10097/0.092634	0.08823/0.111607	0.07729/0.134842
FERD	0.04616/0.077415	0.03353/0.114955	0.02853/0.142248
BSDM	0.03047/0.190544	0.02530/0.290889	0.02317/0.350649
ERX	0.14634/0.203328	0.11673/0.356433	0.09210/0.555702
UADNet	0.03576/0.359679	0.02290/0.558644	0.01614/0.707995
DFAN	0.21013/0.088271	0.16325/0.139002	0.14183/0.179383
RX	0.04349/0.358868	0.03732/0.488433	0.03166/0.618709
LRX	0.00661/0.124696	0.00481/0.224026	0.00371/0.359781
SVM	0.08767/0.619420	0.06771/0.814022	0.04800/0.942370
K-Means	0.19572/0.289671	0.15995/0.412541	0.13130/0.515726
AE	0.01307/0.194095	0.00801/0.339692	0.00607/0.446631
SLRX	0.00435/0.102881	0.00253/0.214996	0.00200/0.286526
MARNet	0.11483/**0.039570**	0.08714/<u>0.061790</u>	0.06981/<u>0.088474</u>
BBO-MARNet	0.10150/<u>0.046165</u>	0.08510/**0.058137**	0.06706/**0.081879**

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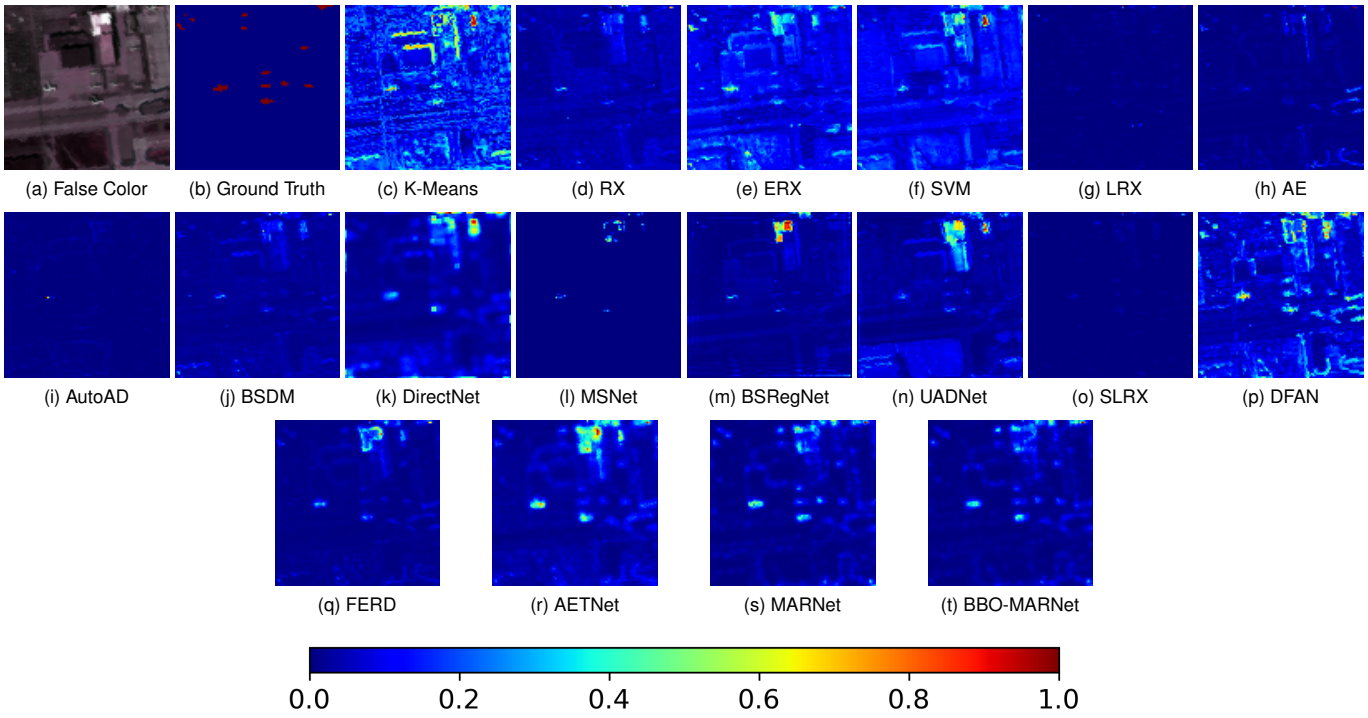


Fig. 7. Detection maps of various methods on the ABU-a dataset. From top to bottom, left to right: False color composite, Ground Truth, K-Means, RX, ERX, SVM, LRX, AE, AutoAD, BSDM, DirectNet, MSNet, BSRegNet, UADNet, SLRX, DFAN, FERD, AETNet, MARNet, and our proposed BBO-MARNet. High anomaly probabilities are highlighted in dark red, while background is suppressed to dark blue.

it further reduces P_F relative to MARNet from 0.061790 to 0.058137 and from 0.088474 to 0.081879. These results indicate that the proposed framework maintains strong background suppression while preserving high target sensitivity on the representative ABU-a scene.

2) Qualitative Analysis: The qualitative detection maps generated by the evaluated methods on the representative datasets (i.e., ABU-a and Ship-53) are visualized in Fig. 7 and Fig. 8. To facilitate an intuitive comparative analysis, all spatial activation maps are normalized to the range of $[0, 1]$ and rendered applying a consistent pseudocolor scheme. In this representation, pixels approaching an intensity of 1 (dark red) indicate a high probability of corresponding to anomalous targets, whereas values near 0 (dark blue) designate suppressed background materials. An optimal hyperspectral anomaly detector should distinctly highlight anomalies with prominent target saliency while systematically attenuating background clutter to minimize the false alarm rate.

As visually evident, traditional statistical and machine learning approaches struggle to handle complex background heterogeneity. Methods such as K-Means, ERX, and SVM fail to adequately characterize intricate background distributions, yielding severe background contamination and elevated false alarm rates (evidenced by the widespread cyan and yellow artifacts within the background regions). Although RX and LRX manage to suppress background interference to a certain degree, they simultaneously attenuate the anomalous signals, resulting in exceptionally low target visibility that renders the anomalies nearly indistinguishable from the local clutter.

Deep learning-based and advanced reconstruction models demonstrate varying degrees of improvement, yet they still face substantial challenges in balancing background suppression with morphological fidelity. Specifically, while AETNet, DirectNet, and FERD successfully localize the general anomalous regions, they suffer from severe spatial blurring and the pervasive *"halo effect"*. This phenomenon causes the detected targets to lose their distinct structural features and artificially dilates their boundaries. Conversely, methods like AutoAD, BSDM, and AE tend to over-smooth the latent representations during the reconstruction phase, significantly degrading the intrinsic target saliency. Furthermore, DFAN exhibits notable vulnerability against structured background clutter, erroneously emphasizing road networks and building edges as anomalies. MSNet and BSRegNet fail to capture the holistic structure of the targets, yielding fragmented or edge-dominated detection responses.

In contrast, the proposed MARNet and its band-optimized variant, BBO-MARNet, achieve the most competitive visual performance across both structured urban landscapes (ABU-a) and highly complex maritime environments (Ship-53). Compared to the aforementioned baselines, BBO-MARNet comprehensively suppresses the heterogeneous background clutter, reducing false alarms to near zero (characterized by the uniformly dark blue background). Concurrently, it rigorously preserves the high-frequency morphological details and sharp boundaries of the anomalies without inducing any halo artifacts. The spatial activation map of BBO-MARNet exhibits the highest fidelity to the ground truth. This definitively demonstrates

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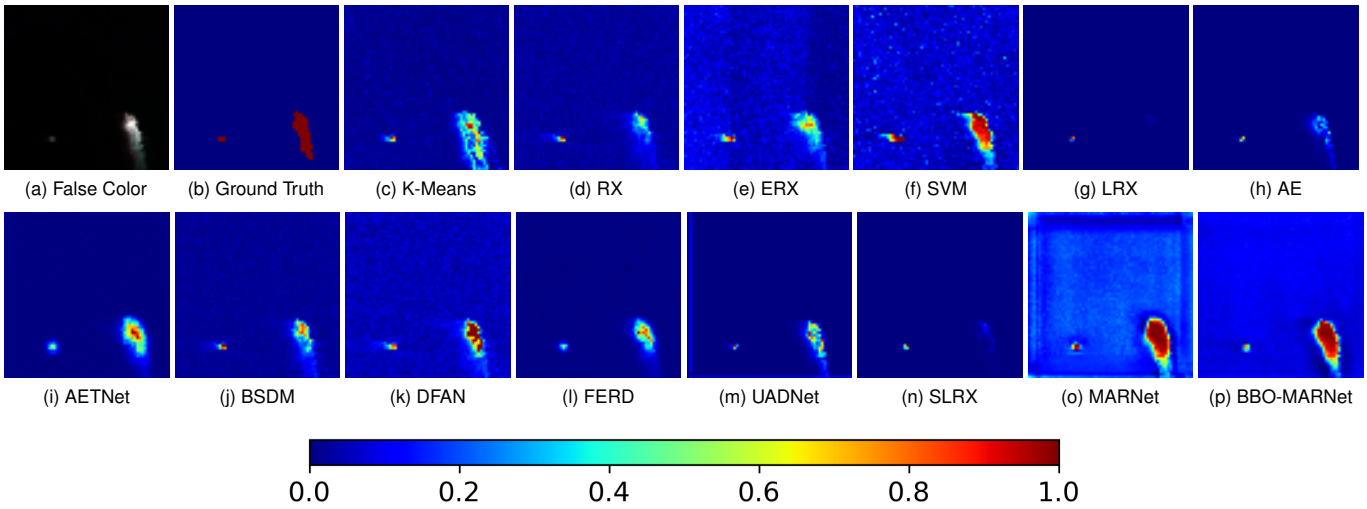


Fig. 8. Detection maps of various methods on the Ship-53 dataset. From top to bottom, left to right: False color composite, Ground Truth, K-Means, RX, ERX, SVM, LRX, AE, AETNet, BSDM, DFAN, FERD, UADNet, SLRX, and our proposed BBO-MARNet. High anomaly probabilities are highlighted in dark red, while background clutters are robustly suppressed to dark blue.

that by effectively extracting the material-sensitive subspace from the extremely redundant hyperspectral space, the target's intrinsic signatures are fully preserved. Concurrently, the conditional mask-aware reconstruction explicitly amplifies the anomaly-background discrepancy. This synergistic integration successfully establishes a highly discriminative and robust manifold boundary, enabling precise target localization even within highly complex maritime or urban clutters.

E. Low-False-Alarm Evaluation and Statistical Analysis

Table II reports the ROC-AUC results following the historical-best checkpoint protocol adopted in the original submission. To provide a complementary and more conservative evaluation, we further recompute the requested metrics using the archived score maps obtained at the terminal epoch of the configured patience-based stopping procedure. No retrospectively selected peak epoch is substituted in this analysis. Therefore, the AUC values reported in this subsection may differ from those in Table II. Importantly, all metrics reported below for a given method and scene are computed from the same terminal score map, ensuring a consistent comparison across ROC-AUC, average precision (AP), maximum F1 score ($F1_{\max}$), and the low-false-alarm measures.

In addition to the full-range ROC-AUC, we evaluate precision ($P_{1\%}$), recall ($R_{1\%}$), and partial AUC ($\mathrm{pAUC}_{1\%}$) under the false-alarm-rate constraint $P_F \leq 1\%$. These complementary metrics provide a more operationally relevant assessment of hyperspectral anomaly detection, where maintaining a low false-alarm rate is important for practical deployment.

Table IV reports all requested metrics for the 18 methods with valid archived terminal score maps on ABU-a. MARNet achieves a marginally higher terminal AUC than BBO-MARNet (0.9731 versus 0.9722), with an absolute difference of only 0.0009. In contrast, BBO-MARNet achieves the highest AP, $F1_{\max}$, $P_{1\%}$, $R_{1\%}$, and $\mathrm{pAUC}_{1\%}$ in this ABU-a comparison.

TABLE IV
DETECTION METRICS ON THE REPRESENTATIVE ABU-A SCENE USING THE ARCHIVED TERMINAL SCORE MAPS. $P_{1\%}$ AND $R_{1\%}$ ARE MEASURED AT $P_F \leq 1\%$. HIGHER IS BETTER; THE BEST AND SECOND-BEST VALUES ARE BOLDFACED AND UNDERLINED.

Method	AUC	AP	$F1_{\max}$	$P_{1\%}$	$R_{1\%}$	$\mathrm{pAUC}_{1\%}$
Auto-AD	0.7342	0.0548	0.0695	0.0745	0.0486	0.0336
BSRegNet	0.7019	0.0513	0.1470	0.0312	0.0208	0.0018
MSNet	0.9260	0.1030	0.2043	0.1548	0.0903	0.0524
DirectNet	0.8797	0.0753	0.1478	0.1429	0.1111	0.0410
AETNet	0.9417	0.1266	0.2296	0.1250	0.0972	0.0430
FERD	0.9460	0.1289	0.2424	0.0938	0.0625	0.0280
BSDM	0.8854	0.0952	0.1860	0.1504	0.1181	0.0476
ERX	0.8802	0.1509	0.2607	0.1966	0.1597	0.1073
UADNet	0.7709	0.0394	0.1037	0.0106	0.0069	0.0004
DFAN	0.9491	0.1871	0.3115	0.2264	0.1667	0.1003
RX	0.8221	0.1012	0.1933	0.2049	0.1736	0.0834
LRX	0.9149	0.1521	0.2527	0.2231	0.1875	0.1115
SVM	0.6990	0.0443	0.1196	0.0600	0.0417	0.0148
KMeans	0.8424	0.0663	0.1614	0.0698	0.0417	0.0277
AE	0.8643	0.0654	0.1390	0.0625	0.0347	0.0167
SLRX	0.9285	0.2200	0.3091	0.2950	0.2847	0.1827
MARNet	**0.9731**	<u>0.3139</u>	<u>0.3989</u>	<u>0.3660</u>	<u>0.3889</u>	<u>0.2244</u>
BBO-MARNet	<u>0.9722</u>	**0.3342**	**0.4000**	**0.3806**	**0.4097**	**0.2697**

In particular, $\mathrm{pAUC}_{1\%}$ increases from 0.2244 for MARNet to 0.2697 for BBO-MARNet, while $R_{1\%}$ increases from 0.3889 to 0.4097. These results indicate that task-aware band optimization can improve the precision-recall and low-false-alarm operating characteristics even when the difference in full-range ROC-AUC is negligible.

To further isolate the contribution of Bayesian band optimization, Table V directly compares BBO-MARNet with its MARNet backbone across all evaluated datasets. Each entry represents the difference between BBO-MARNet and MARNet using the same terminal-checkpoint evaluation protocol.

The cross-dataset comparison in Table V shows that the

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TABLE V
INCREMENTAL EFFECT OF BAYESIAN BAND OPTIMIZATION ON COMPLEMENTARY PRECISION-RECALL AND LOW-FALSE-ALARM METRICS RELATIVE TO MARNET USING THE ARCHIVED TERMINAL CHECKPOINTS. EACH ENTRY REPORTS BBO-MARNET MINUS MARNET; POSITIVE VALUES INDICATE IMPROVEMENTS.

Dataset	ΔAP	$\Delta F1_{\max}$	$\Delta P_{1\%}$	$\Delta R_{1\%}$	$\Delta \mathrm{pAUC}_{1\%}$
San Diego	+0.0077	+0.0700	+0.0547	+0.1552	+0.0267
ABU-a	+0.0203	+0.0011	+0.0146	+0.0208	+0.0454
ABU-b	+0.2732	+0.2696	+0.0250	+0.3158	+0.3406
ABU-u	+0.0708	+0.0951	+0.0991	+0.1154	+0.1136
Coast	-0.1224	-0.0705	-0.0269	-0.1493	-0.1021
Ship	-0.0238	-0.0285	-0.0515	-0.0010	-0.0223
HAD100	+0.0025	+0.0010	-0.0072	+0.0006	0.0000

effect of Bayesian band optimization is both scene- and metric-dependent. BBO-MARNet improves all five complementary precision-recall and low-false-alarm metrics on San Diego, ABU-a, ABU-b, and ABU-u. The largest gains are observed on ABU-b, where AP, $F1_{\max}$, $R_{1\%}$, and $\mathrm{pAUC}_{1\%}$ increase by 0.2732, 0.2696, 0.3158, and 0.3406, respectively. On San Diego, $F1_{\max}$ and $R_{1\%}$ increase by 0.0700 and 0.1552, respectively, further demonstrating the improvement in low-false-alarm detection performance.

The improvements are not universal across all scenes. MARNet performs better on most complementary metrics on Coast and Ship, whereas the differences on HAD100 are comparatively small. Therefore, the terminal-checkpoint results do not support a claim that Bayesian band optimization uniformly improves every metric on every dataset. Instead, they indicate that task-aware spectral subspace optimization can provide substantial improvements in precision-recall and low-false-alarm performance on several challenging scenes while maintaining broadly comparable full-range AUC.

Statistical significance was further assessed using AUC as a common prespecified statistic. For each single-scene benchmark, BBO-MARNet was compared with both the external method having the highest terminal AUC and the MARNet backbone using paired spatial-block resampling. Specifically, paired 10×10-pixel spatial blocks were used to account for local spatial dependence, 2,000 bootstrap resamples were used to estimate pointwise 95% confidence intervals, and two-sided paired block-permutation tests were used for hypothesis testing. For the multi-scene Ship and HAD100 datasets, scene-level AUC differences were evaluated using two-sided paired Wilcoxon signed-rank tests together with 2,000-resample paired scene-level bootstrap confidence intervals. Holm correction was applied across all 14 comparisons to control the family-wise error rate.

After correction for multiple comparisons, none of the 14 AUC comparisons reached statistical significance ($p_{\mathrm{Holm}} \geq 0.0957$). Accordingly, we do not claim universal or statistically significant superiority of BBO-MARNet in terms of full-range ROC-AUC. The pointwise bootstrap confidence intervals and Holm-adjusted hypothesis tests address different inferential quantities, so exclusion of zero from an unadjusted pointwise confidence interval does not necessarily imply significance after correction for multiple comparisons. The complete statistical test results are provided in Table VIII in the supplementary material. Overall, these results support a conservative interpretation that the principal benefit of BBO-MARNet lies in improved precision-recall and low-false-alarm operating trade-offs on several datasets rather than in a universally significant increase in full-range ROC-AUC.

F. Ablation and Analytical Studies

1) Effectiveness of Composite Loss Components: To explicitly decouple the influence of the band selection module and rigorously isolate the individual contributions of the loss constraints, all ablation experiments in this subsection are consistently conducted using a fixed subset of the first 50 spectral bands. The proposed optimization framework comprises three progressive components: the gradient structural loss ($\mathcal{L}_{grad}$), the multi-scale background reconstruction loss ($\mathcal{L}_{bg}$), and the internal anomaly regularization loss ($\mathcal{L}_{in}$). The quantitative impacts of these individual components and their various combinations on the mean AUC are comprehensively detailed in Table VI.

TABLE VI
ABLATION STUDY RESULTS WITH DIFFERENT LOSS COMPONENTS

$\mathcal{L}_{grad}$	$\mathcal{L}_{bg}$	$\mathcal{L}_{in}$	**Mean AUC**
✓			0.99275
	✓		0.99406
		✓	0.98794
✓	✓		0.99343
✓		✓	0.98896
	✓	✓	<u>0.99424</u>
✓	✓	✓	**0.99431**

When deployed independently, $\mathcal{L}_{bg}$ yields the most substantial performance gain, achieving a mean AUC of 0.99406. This empirical observation underscores that robust, multi-scale background modeling remains the most critical driver for successful hyperspectral anomaly detection. Conversely, relying exclusively on $\mathcal{L}_{in}$ results in the lowest AUC (0.98794). Since $\mathcal{L}_{in}$ solely penalizes reconstruction errors within the anticipated anomalous regions without incorporating global contextual constraints, the network suffers from severe representation collapse, fundamentally failing to characterize the complex background clutter.

The integration of paired loss functions reveals intricate optimization dynamics. The combination of $\mathcal{L}_{bg}$ and $\mathcal{L}_{in}$ yields a highly competitive AUC of 0.99424. This dual-constraint mechanism systematically compels the network to maintain high-fidelity background reconstruction while actively suppressing the reconstruction of anomalous targets, thereby maximizing the residual gap between anomalies and the background. Notably, coupling $\mathcal{L}_{grad}$ strictly with $\mathcal{L}_{bg}$ slightly degrades the performance (0.99343) compared to utilizing $\mathcal{L}_{bg}$ alone. This implies that enforcing strict structural gradient similarity over the entire image—without explicitly restricting the anomaly representations via $\mathcal{L}_{in}$—inadvertently coerces the network into preserving anomalous textures, leading to a subtle reduction in target detectability.

The proposed composite objective function, which seamlessly integrates all three components, achieves the optimal

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mean AUC of 0.99431. In this synergistic configuration, the components fulfill distinct yet complementary roles: $\mathcal{L}_{bg}$ guarantees global background purification, $\mathcal{L}_{in}$ ensures targeted anomaly suppression, and $\mathcal{L}_{grad}$ meticulously preserves morphological boundaries and spatial high-frequency details. This tripartite optimization effectively mitigates the spatial blurring and "halo effects" commonly observed in traditional reconstruction-based models, establishing a robust and highly discriminative latent space for anomaly separation.

2) Effectiveness of Task-Aware Band Selection and Detection Strategies: To rigorously validate the efficacy of the proposed Task-Aware Bayesian Band Selection module and the mask-guided reconstruction strategy, comprehensive ablation studies were conducted. We juxtaposed our Bayesian selection mechanism against a baseline strategy that statically selects the first K contiguous bands—a configuration notoriously susceptible to severe spectral collinearity. Experiments were executed under strict dimensionality constraints ($K \in \{8, 16, 32, 64\}$). Furthermore, to identify the optimal anomaly scoring mechanism, four distinct detection protocols were evaluated based on the network's input and output tensors: *Origin* (detection executed directly on the selected input sub-cube), *Recon* (detection on the reconstructed output), *Add* (detection based on the element-wise sum of the input and reconstructed output), and *Dif* (detection based on the absolute difference between the input and output). The quantitative results, averaged over 8 independent runs and measured by the Mean AUC (mAUC), are detailed in Table VII.

TABLE VII
ABLATION STUDY ON BAND SELECTION STRATEGIES AND DETECTION METHODS

Bands (K)	Selection Strategy	Origin	Recon	Add	Dif	Best mAUC
8	First K (Baseline)	0.93490	0.97024	0.97130	0.96837	0.97156
	Bayesian (Ours)	**0.98353**	**0.98105**	**0.98486**	**0.97054**	**0.98524**
16	First K (Baseline)	0.96077	0.98212	0.98289	0.97971	0.98290
	Bayesian (Ours)	**0.98379**	**0.98865**	**0.99056**	**0.98351**	**0.99056**
32	First K (Baseline)	0.97705	0.99244	0.99214	0.99096	0.99244
	Bayesian (Ours)	**0.98242**	**0.99337**	**0.99333**	**0.99191**	**0.99338**
64	First K (Baseline)	**0.97784**	**0.99336**	**0.99304**	**0.98998**	**0.99336**
	Bayesian (Ours)	0.97767	0.99305	0.99274	0.98974	0.99305

Several critical observations can be drawn from the empirical results presented in Table VII:

1) Benefits and limits of task-aware band selection: The Bayesian band optimization provides clear advantages under compact spectral budgets. When restricted to 8 bands, the Bayesian selection achieves a peak mAUC of 0.98524, far exceeding the baseline's 0.97156. Similarly, the 16-band Bayesian setting (0.99056) nearly rivals the 32-band baseline (0.99244), confirming that BBO isolates highly discriminative spectral channels under strong dimensionality constraints. However, when $K = 64$, the First-K baseline is slightly better than Bayesian selection in the best-score column (0.99336 versus 0.99305). This small reversal indicates an applicability limit rather than a failure of the overall framework: as K increases, the retained subspace already contains sufficient discriminative information, the marginal benefit of discrete search diminishes, and the BO objective may overfit small calibration-scene fluctuations. Therefore, BBO is most valuable for aggressive compression and deployment-oriented settings, while large-K configurations should be selected by validation or replaced by simpler band retention when computation is not constrained.

2) Efficacy of Mask-Guided Reconstruction: Across the majority of configurations, the deep reconstruction strategies (i.e., *Recon* and *Add*) yield substantial performance gains over detection performed on the raw *Origin* data. For example, under the baseline selection at 16 bands, the *Add* protocol elevates the mAUC from 0.96077 (*Origin*) to 0.98289. This validates the core premise of MARNet: the mask-aware self-supervised reconstruction robustly purifies the background manifold, inherently amplifying target-background separability prior to the final detection phase.

3) Optimal Scoring Mechanism and Early Performance Saturation: Among the evaluated scoring protocols, the *Add* and *Recon* strategies predominantly deliver the most competitive and stable detection accuracies across varying spectral subset sizes. Crucially, while both selection strategies achieve exceptional performance as the band count expands to 64 (mAUC > 0.993), the proposed Bayesian method reaches its performance plateau earlier, peaking at 32 bands (mAUC of 0.99338). The $K = 64$ result may be related to performance saturation and redundancy. When more bands are retained, the marginal benefit of task-aware selection may diminish, and additional bands could reduce the distinguishability gained from selection. This early saturation highlights that BBO-MARNet provides a favorable trade-off between computational overhead and detection accuracy, requiring fewer than half the spectral bands to attain state-of-the-art anomaly localization.

3) Feature Space Visualization: To qualitatively validate the efficacy of the proposed task-aware Bayesian band selection strategy, we visualize the feature space distributions of the Texas Coast dataset before and after dimensionality reduction. Figure 9 presents the 2D embeddings of the background and anomalous target pixels utilizing both t-SNE and UMAP manifold learning algorithms.

When utilizing the full spectrum of 204 bands (Fig. 9(a) and (c)), the high-dimensional feature space is severely corrupted by spectral redundancy and inter-band noise. Although the anomalous targets (labeled in orange) exhibit a loose degree of aggregation, they remain inextricably entangled with the periphery of the background manifold (labeled in blue). This topological overlap practically illustrates the Hughes phenomenon; excessive uninformative bands obscure the intrinsic discriminative features, establishing a highly convoluted decision boundary that hinders downstream anomaly detectors.

In stark contrast, after applying the Bayesian optimization module to isolate the 15 most informative bands (Fig. 9(b) and (d)), the topological structure of the latent space undergoes a profound transformation. Despite a substantial compression in dimensionality, the target pixels are embedded into much tighter, highly concentrated clusters that are explicitly repelled from the dominant background distribution. This disentanglement is particularly evident in the UMAP visualization (Fig. 9(d)), where the selected 15-band subset projects the targets into an isolated manifold with negligible background overlap.

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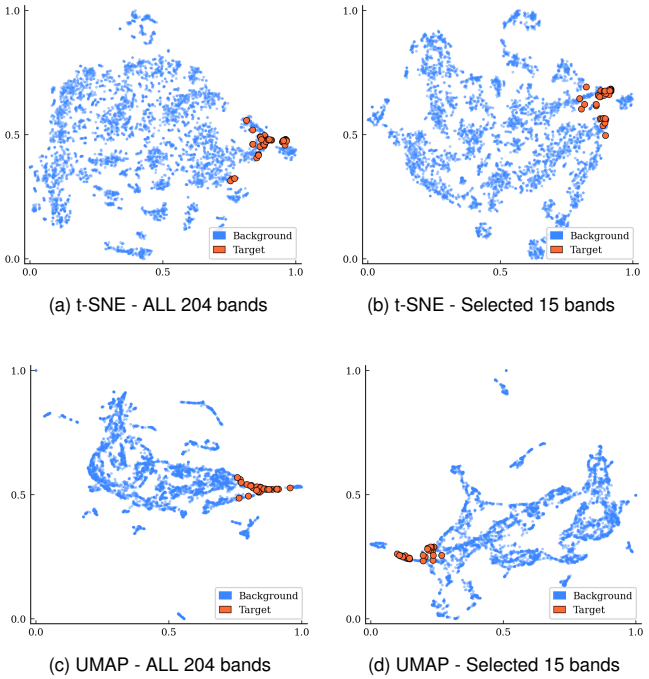


Fig. 9. Visualization of feature space distributions on the Texas Coast dataset before and after task-aware band selection. (a) and (b) illustrate the local structure via t-SNE, while (c) and (d) depict the global topological manifold via UMAP.

This visualization provides compelling empirical evidence that the proposed task-aware band selection transcends naive redundancy reduction. By actively optimizing for downstream detection metrics, BBO-MARNet successfully projects the hyperspectral data into a highly discriminative subspace. This fundamental enhancement of intrinsic target-background separability decisively alleviates the representational burden on the subsequent mask-guided reconstruction network, thereby serving as a critical catalyst for the overall state-of-the-art detection accuracy.

V. Conclusion

This work presents BBO-MARNet, a compact yet highly effective framework for hyperspectral anomaly detection, designed to address two tightly coupled bottlenecks in existing reconstruction-based paradigms: severe spectral redundancy and the identity mapping tendency of deep generative models. The proposed approach integrates a task-aware Bayesian band optimization strategy to identify a discriminative, material-sensitive subspace, together with a mask-guided spatial–spectral Transformer that enforces context-consistent background reconstruction while explicitly suppressing anomaly memorization. To further enhance reconstruction fidelity and discrimination capability, a tripartite structure-preserving loss is introduced to jointly regulate spectral consistency, spatial coherence, and anomaly–background separability, thereby amplifying residual contrast in a principled manner. Extensive experiments on five benchmark datasets, including the large-scale HAD100 dataset and a newly constructed challenging maritime Ship dataset, demonstrate that BBO-MARNet consistently achieves state-of-the-art performance in anomaly detection accuracy. Notably, the proposed framework contains only 1.69 million parameters and incorporates a cross-scene band transfer mechanism, enabling cross-scene generalization to unseen target scenes without using target-scene labels. These properties make BBO-MARNet particularly suitable for resource-constrained, on-board remote sensing scenarios, where both efficiency and robustness are critical.

Despite these advantages, three limitations remain. First, the Bayesian band optimization relies on offline discrete search, introducing non-negligible computational overhead. Second, the current stochastic masking strategy is content-agnostic and may not fully capture semantic variations in highly heterogeneous backgrounds. Third, when AUC is used as the BBO feedback signal, a labeled calibration or reference scene is required; deployment without a labeled calibration/reference scene would require replacing this objective with a proxy that does not require calibration labels. Future work will focus on developing differentiable or unsupervised band selection schemes to enable fully end-to-end optimization, as well as adaptive saliency-aware masking mechanisms. Extending the framework to multi-temporal and hyperspectral video data streams also represents a promising direction toward real-time dynamic Earth observation.

Appendix

TABLE VIII
PAIRED AUC TESTS BASED ON THE ARCHIVED TERMINAL SCORE MAPS. BBO-MARNET IS COMPARED WITH THE STRONGEST EXTERNAL BASELINE AND THE MARNET BACKBONE. CIS ARE POINTWISE 95%; HOLM-ADJUSTED p VALUES CONTROL THE FAMILY-WISE ERROR RATE ACROSS THE 14 COMPARISONS.

Dataset	Comparator	ΔAUC [95% CI]	p_{Holm}
San Diego	ERX	-0.0129 [-0.0364, +0.0073]	1.0000
San Diego	MARNet	-0.0107 [-0.0271, -0.0005]	0.6477
ABU-a	DFAN	+0.0230 [+0.0048, +0.0487]	0.2339
ABU-a	MARNet	-0.0010 [-0.0052, +0.0026]	1.0000
ABU-b	DFAN	-0.0015 [-0.0080, +0.0144]	1.0000
ABU-b	MARNet	+0.0072 [+0.0028, +0.0120]	1.0000
ABU-u	BSRegNet	-0.0372 [-0.0652, -0.0135]	0.1259
ABU-u	MARNet	-0.0086 [-0.0314, +0.0146]	1.0000
Coast	MSNet	-0.0112 [-0.0210, -0.0032]	0.1169
Coast	MARNet	-0.0113 [-0.0217, -0.0024]	0.1799
Ship	BSDM	-0.0017 [-0.0042, -0.0002]	0.0957
Ship	MARNet	-0.0004 [-0.0009, +0.0000]	1.0000
HAD100	FERD	+0.0020 [+0.0006, +0.0036]	0.1555
HAD100	MARNet	-0.0002 [-0.0006, +0.0001]	1.0000

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